



UNIVERSITY OF
LIVERPOOL

Application Development with C++ (ELEC362)

Lecture 1: Introduction

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This lecture

- What is covered in this lecture?
 1. Introduction to the module.
 2. Overview of C++.
- Why is it covered?
 1. To explain the module's structure and assessment.
 2. To appreciate the advantages of C++ .

The learning outcomes

- This module will teach you:
 1. How to use C++ to implement console-based software.
 2. How to use C++ to implement Graphic User Interface (GUI)-based software.
 3. How to effectively use online documentations and resources for self-learning in programming context.

Note: This module is NOT a computer science degree in a single module!

Basic information

- Pre-request modules:

ELEC129

Or ELEC102 (XJTLU students)

Or COMP122 (Comp Sci & EE students)

- Background knowledge : algorithms, basic programming experience, and basic maths.
- Statistics from last year: Average mark (61%) Failure rate (6.6%) Top mark (84).

Workload and assessment

- The module is 15 credits module (corresponding to 150 hours of study) distributed as:

Lectures (18 hours)

Lab sessions (36 hours)

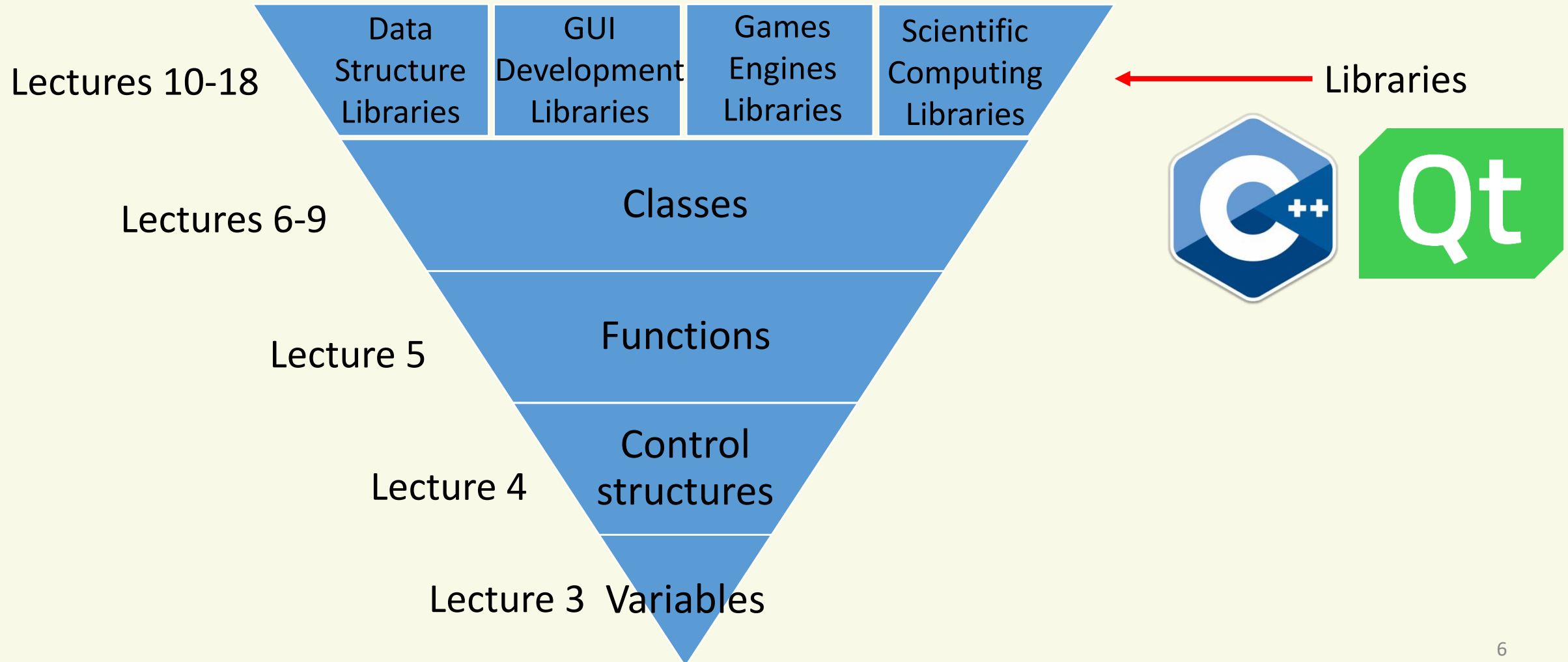
Project and assignment (55 hours total)

Self-study (39 hours)

- Mark distribution:

Assessment	Weight (%)	Assessment aspect	Deadline
Assignment	30	Console-based software	Week 6
Project	70	GUI-based software Independent learning	Week 13

Topics covered



Resources

- Books: Professional C++ , Marc Gregoire (2018).
C++ GUI Programming with Qt, J Blanchette & M Summerfield (2008).
- Online documentation: C++ : <http://www.cplusplus.com/reference/>
Qt : <https://doc.qt.io/>
- Educational websites: <https://www.geeksforgeeks.org/c-plus-plus/>
<https://www.tutorialspoint.com/cplusplus/>
- Online developers community: <https://stackoverflow.com/>
<https://forum.qt.io/>

C++ Overview

- C++ is a general purpose programming language, appeared in 1983.
- Since its creation, many standard editions of C++ where released including C++ 98, C++ 11, C++ 14, and C++ 17.

- In its early times there were many other programming languages:

Language	Primary use
COBOL	Business theme
Fortran	Scientific computing
Pascal	Data structures

- The Object Oriented Programming (OOP) paradigm enabled C++ to be customisable for any field.

C++ Overview

- As a result, C++ became widely used on all levels of software including:

Operating systems



Applications



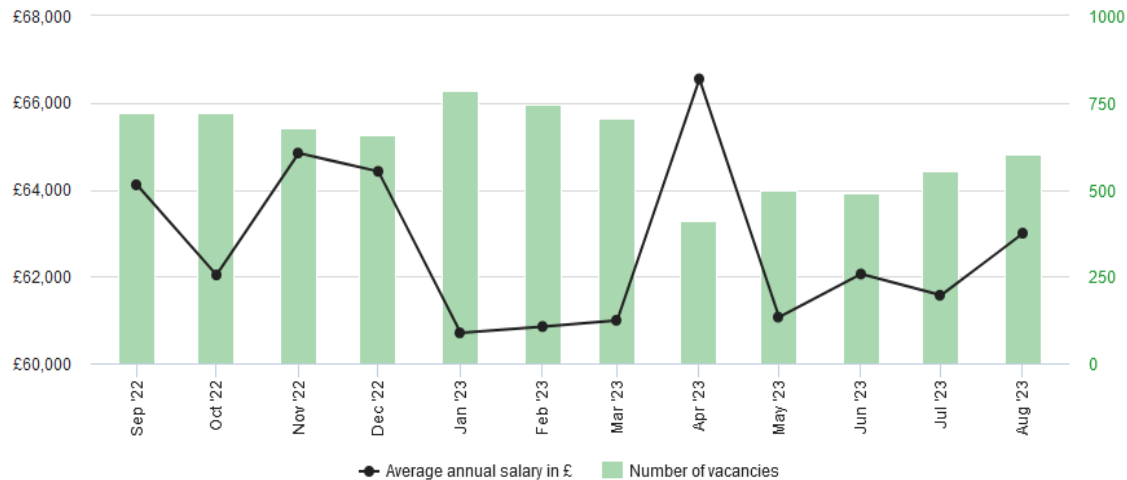
Firmware



C++ Overview

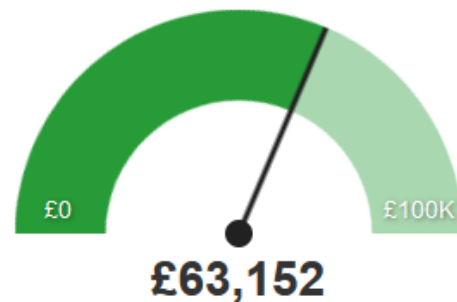
- Despite it being old, C++ is still a valuable skill to have:

Salary and vacancy trends for C++ engineer, UK

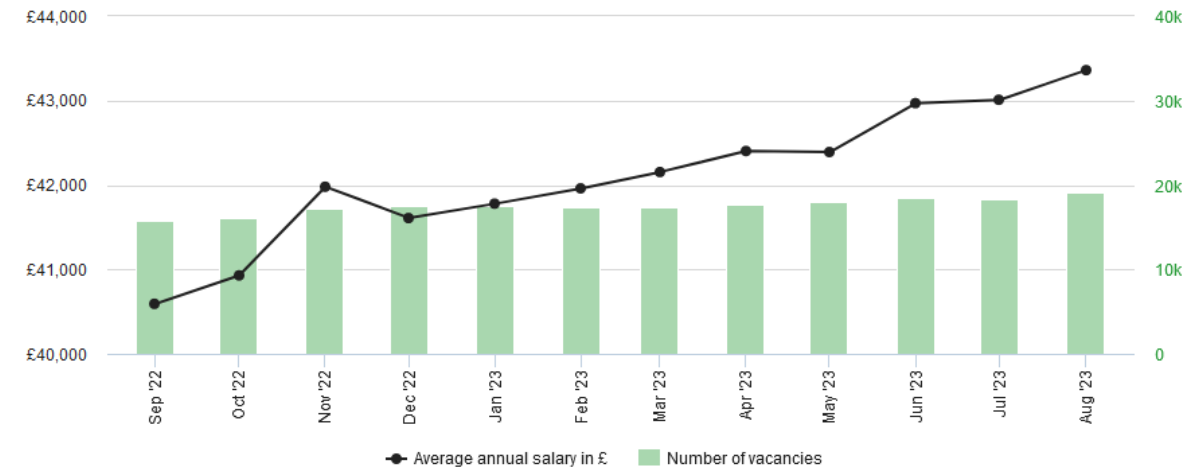


<https://www.adzuna.co.uk/jobs/salaries/c++-engineer>

Average current salary

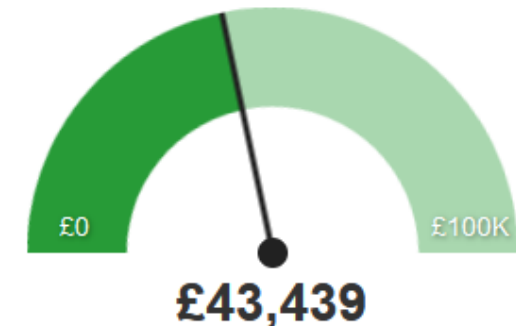


Salary and vacancy trends for Electrical engineering, UK



<https://www.adzuna.co.uk/jobs/salaries/electrical-engineering>

Average current salary



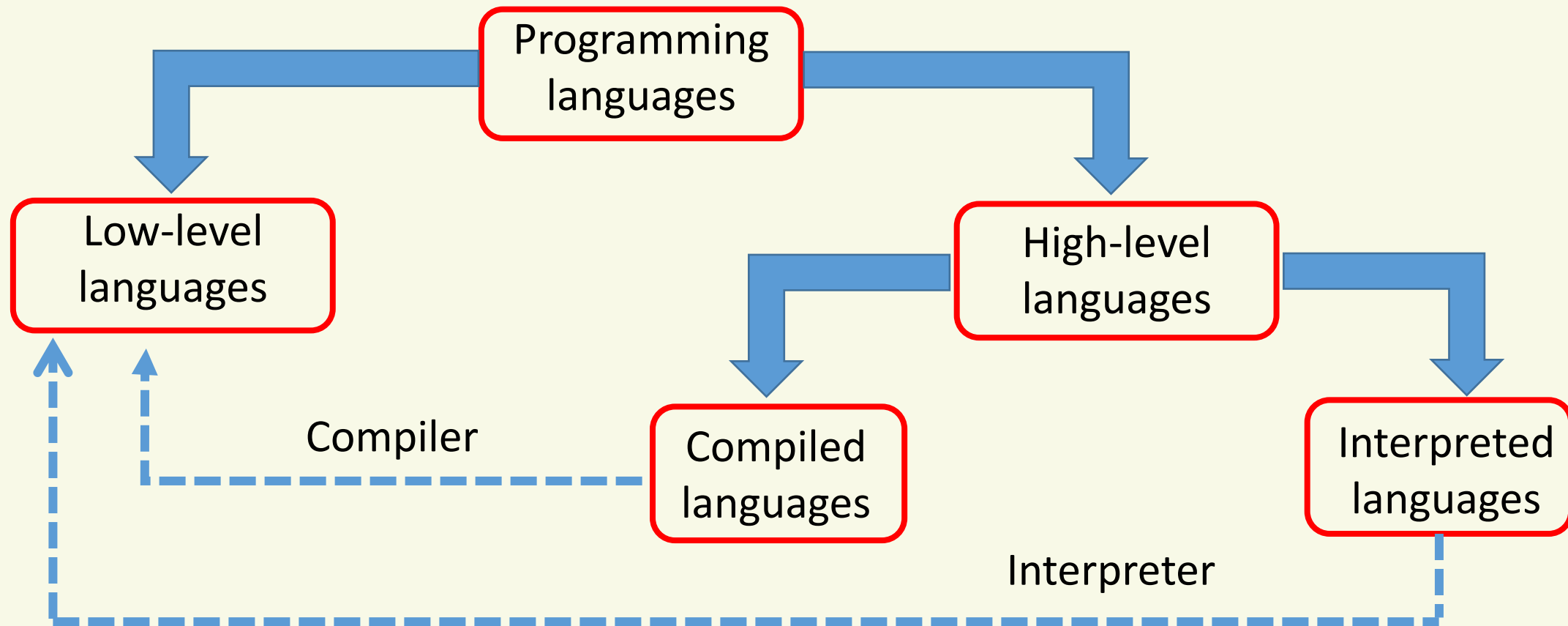
High level programming languages

- Based on the level of programming languages, they are split into two categories:
 1. High level languages: programming languages that use human readable syntax.

Examples include: C++, MATLAB, Python.
 2. Low level languages: programming languages that use machine readable syntax.

Examples include: Assembly language.
- The computer only understands Low level languages.
- We need a “translator” software to translate high level to low level languages.

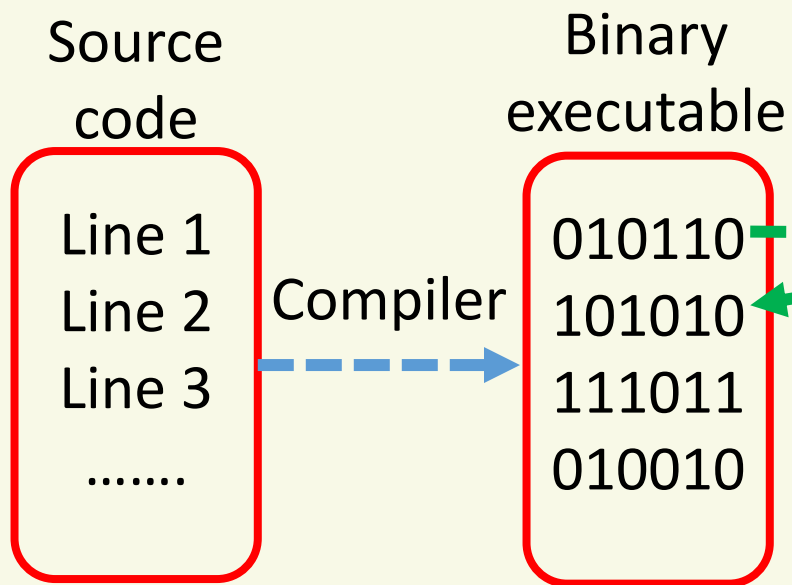
High level programming languages



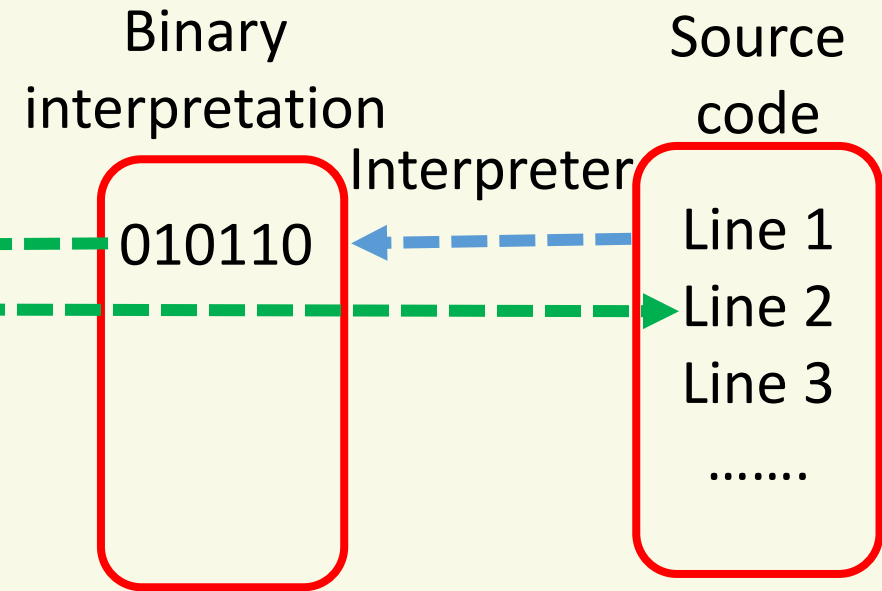
- The “translator” can be either a compiler or an interpreter, thus leading to compiled languages and interpreted languages.

Compiled vs Interpreted languages

- Compiled languages:



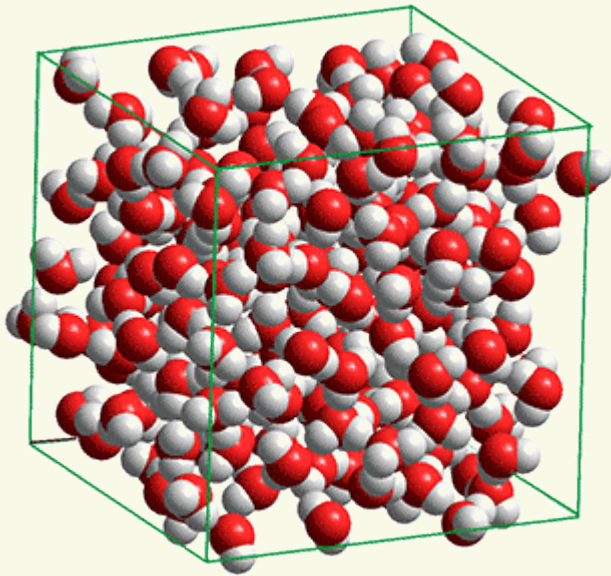
- Interpreted languages:



- QUESTION: Based on this explanation, are there any clear advantages of either languages?

Compiled vs Interpreted languages

- Example of compiled languages are C++ and Fortran. While examples of interpreted languages are MATLAB and Python.



Simulation steps	Time (Fortran)	Time (MATLAB)
7000	15 sec	4439 sec
25000	38 sec	13 hours
105000	138 sec	???

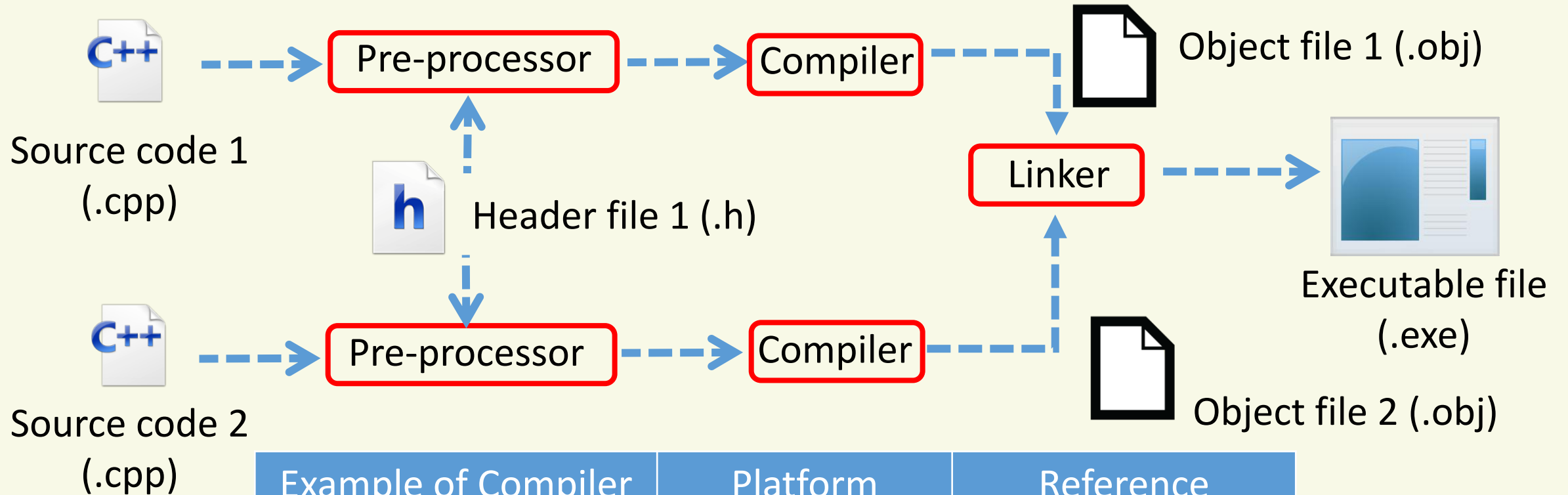
http://utkstair.org/clausius/docs/che548/pdf/md_sim.pdf

- QUESTION: Is it all bad for interpreted languages?

The compilation process

- The code written by the software developer is known as the source code.
- Source code: The version of software as it is originally written by a human in plain text.
- Header file: A file containing parts of code to be shared among multiple source files.
- Executable file: A file that runs the programme when it is opened.
- The end user only deals with the executable file.

The compilation process



Example of Compiler	Platform	Reference
MSVC	Windows	link
MinGW	Windows	link
g++	Linux	link

Integrated Development Environments (IDE)

- Code development is done using software known as IDE.
- For this module we will use Visual Studio (community version):



<https://visualstudio.microsoft.com/downloads/>

- Visual studio is standardly used in the job market.
- Knowing it will boost your CV!!

Software Engineer - Software Developer

Software Engineer / Developer with experience in one or more of the following technologies: Windows / Linux / Solaris Platforms; C++; Java / JavaScript; C#; MySQL; Big Data Secure Solutions; Qt; Visual Studio / Eclipse; Agile Development or JIRA / Atlassian Toolset is required for a leading...

Date: 26th April Location: Lincoln, Lincolnshire, East Midlands Salary/Rate: up to £50,000 per annum + Benefits

More Links

View Job

Add to Basket

ISL

C++ Software Developer/ Engineer - MS Visual Studio - Manchester

🕒 Apply By: 14th April

👍 APPLY DIRECTLY

📄 VIEW MORE @ ISL

Role: C++ Software Engineer/ Developer – MS Visual Studio

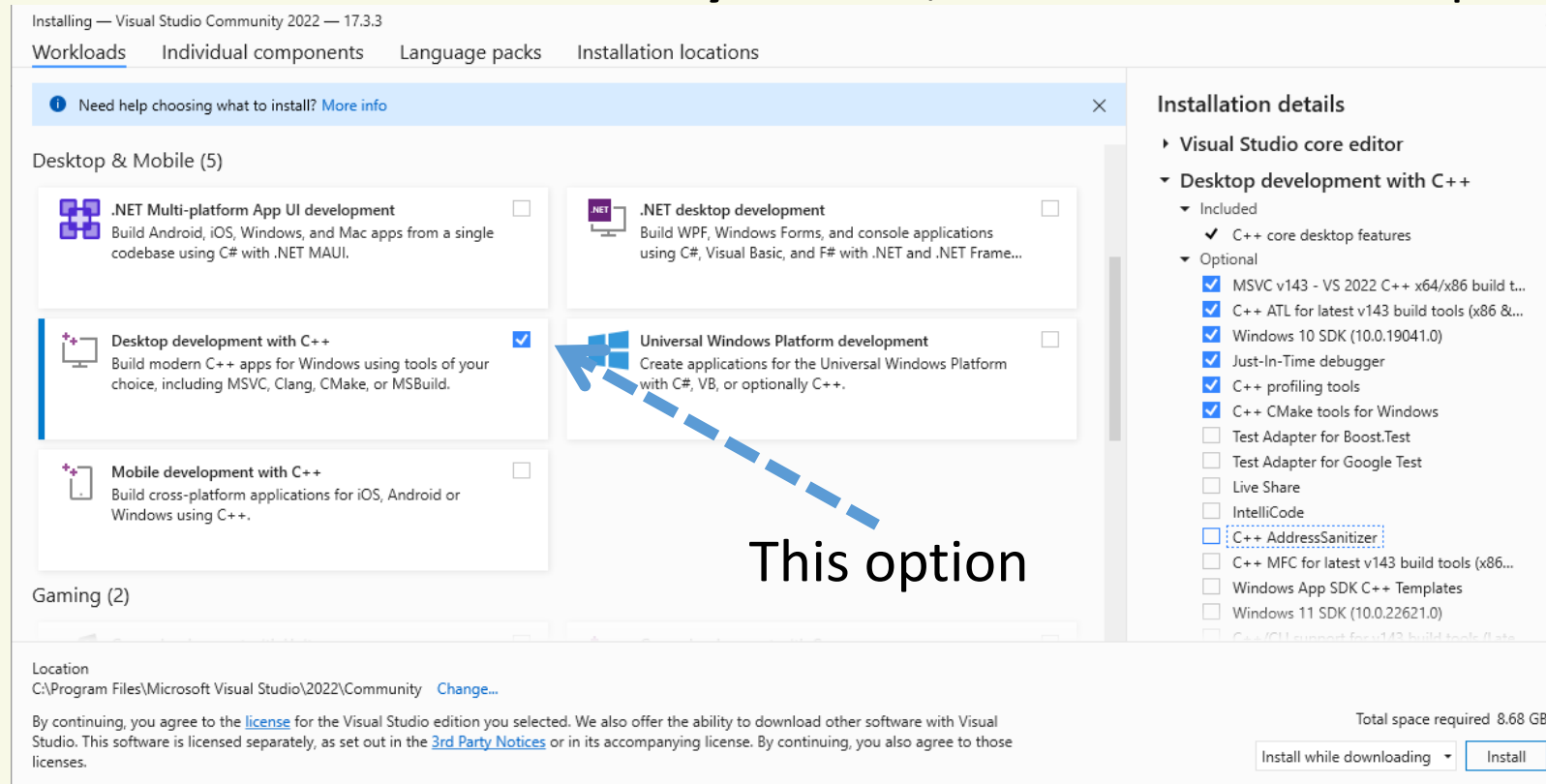
Location: Commutable from – Central Manchester + 1hour

Salary/click apply There is a senior and mid-level position available)

Key Skills: C++, MS Visual Studio, Computer science, engineering

Installing Visual Studio

- Download Visual Studio from <https://visualstudio.microsoft.com/downloads/>
- Download the **community** version, choose the follow option:



- Login within 30 days (only on your PC or laptop, not on Uni's computers).
- MAC users can choose any IDE (XCode is a popular choice)

Basic rules of a C++ code

- Go to [L1D1.cpp](#)
- Every programme consists of a set of statements.
- Every line has to end with a semicolon;
- The code consists of one or more blocks (a block starts with { and ends with }).
- Every programme must have one and only one main block/function.
- For comments, // is used to make a line of comment, while /* and */ are used to mark the beginning and the end of a block of comments.

Commenting codes

- Comments are explanatory statements written by the developer in the code to make it more readable by other developers.
- Comments are very important particularly in projects with multiple developers.
- Summary for good practice of coding can be found [here](#).

Practical notes:

- Follow the comment convention when working on a project with multiple developers.
- Convention for this module: state the functionality of every code block in a comment, in addition to commenting any statement with a new function.

Summary

- An overview of C++
- Difference between compiled and interpreted languages.
- The role of compilers and how executable files are created.
- Introduction to Visual Studio.
- Basic rules of C++ coding.